

Compact operators lie in a singleton representation range

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_of_compact_singleton`

theorem

Obtains the inclusion of all compact operators in the image of the original algebra.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed, let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ be a $*$ -representation representing the unique unitary-equivalence class of nonzero irreducible $*$ -representations of A . If $T \in \mathcal{B}(H)$ is compact, then

$$\exists a \in A, \quad \pi(a) = T.$$

Thus $\mathcal{K}(H) \subseteq \pi(A)$.

Assumptions

Separability is assumed for H , not for A . The singleton hypothesis includes nonzero irreducibility of π and the universal unitary-equivalence condition. Faithfulness is not an input to this theorem.

Conclusion

The preimage belongs to A itself. The reverse inclusion $\pi(A) \subseteq \mathcal{K}(H)$ is a separate result.

Proof route

Use faithfulness to close the range, put every rank-one operator in it, and approximate an arbitrary compact operator by finite sums of rank-one operators.

▼ Proof steps

1. Use the inner product linear in its second argument. Apply Faithfulness of a non-unital singleton model to this nonzero ordered A and this singleton model π to obtain injectivity. An injective C^* -homomorphism is isometric, so completeness of A makes $\pi(A)$ norm closed. Apply Every rank-one operator has a preimage to the same π , using separability of H , to obtain preimages of every operator $\theta_{x,y}(z) = \langle y, z \rangle x$. Additivity then supplies preimages of their finite sums.
2. Here is the compact approximation in Compact approximation inside the closed representation range. For $\varepsilon > 0$, the set $K = \overline{T(\overline{B_H(0,1)})}$ is compact, so choose a finite $\varepsilon/2$ -net N for it. Let $E = \text{span}_{\mathbb{C}} N$ and let P_E be its orthogonal projection. Since E is finite dimensional, P_E exists and minimizes distance to E . For

$\|z\| \leq 1$, choose $n \in N$ with $\|Tz - n\| < \varepsilon/2$; then

$$\|Tz - P_ETz\| \leq \|Tz - n\| < \varepsilon/2.$$

Taking the operator norm gives $\|T - P_ET\| \leq \varepsilon/2 < \varepsilon$.

3. Choose an orthonormal basis $e_1, \dots, e_m$ of E . By the inner-product convention just fixed,

$$P_ET(z) = \sum_{j=1}^m \langle e_j, Tz \rangle e_j = \sum_{j=1}^m \langle T^*e_j, z \rangle e_j = \sum_{j=1}^m \theta_{e_j, T^*e_j}(z).$$

Step 1 puts this finite sum in $\pi(A)$; the empty sum when $E = 0$ is also allowed. Arbitrarily close such approximants place T in the closure of $\pi(A)$, which equals $\pi(A)$. Unpacking membership gives the required a .

Main citations

- Exact theorem and proof
- Faithfulness of a non-unital singleton model
- Every rank-one operator has a preimage
- Compact approximation inside the closed representation range

Lean source signature (exact)

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theorem exists_preimage_of_compact_singleton [Nontrivial A]
  [PartialOrder A] [StarOrderedRing A]
  [TopologicalSpace.SeparableSpace H]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)
  (T : H →L[ℂ] H) (hT : IsCompactOperator T) :
  ∃ a : A, pi a = T
```

In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[PartialOrder A] [StarOrderedRing A]	These typeclasses provide the usual order on a C^* -algebra, equivalently $a \leq b$ when $b - a$ is positive. They are used by the state and positivity arguments.
[TopologicalSpace.SeparableSpace H]	The Hilbert space H has a countable dense subset; no separability of A is imposed.
pi : NonUnitalCStarRepresentation A H	The $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$, with no unit-preservation requirement.

In the source	Mathematical meaning
<code>hsingle : IsSingletonIrreducibleModel.{u, v, u} pi</code>	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
<code>(T : H →L[ℂ] H) (hT : IsCompactOperator T)</code>	A specified bounded complex-linear operator $T : H \rightarrow H$ that maps bounded sets to relatively compact sets.
<code>∃ a : A, pi a = T</code>	For this T there is an element a of the original algebra whose represented operator equals T on all of H .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_of_compact_singleton`

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